

Kinematic Restitution Balancing (KRB)

A Low-Cost Analytical Fix for Energy Gain in Velocity-Level Impulse Solvers

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Abstract

In discrete physics simulations using velocity-level impulse solvers (Sequential Impulse / Projected Gauss-Seidel), energy gain is a common numerical artifact. This paper introduces **Kinematic Restitution Balancing (KRB)**, a per-constraint correction for that leak in both unilateral constraints (collisions) and bilateral constraints (joints). By compensating for force-induced velocity drift (Component A) and auditing potential-energy shifts during position correction (Component B), KRB removes the two usual analytic energy sources inside a single-pass SI/PGS pipeline, without a second position pass. It does not claim a global energy invariant; later sections state the coupled-constraint gap that follows from the per-constraint audit.

1. Introduction

Technique, implementation, evaluation, and limitations are deferred to later revisions of this manuscript.

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Code and Data License

Supplemental source code and data referenced in later revisions of this article are distributed under the ISC License (Open Source, BSD-class); see `LICENSE` in the

Gearbox2D repository (<https://github.com/JSideris/Gearbox2D>).

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